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CEO entrepreneurial orientation, human resource management systems, and employee innovative behavior: An attention-based view

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Abstract

Research Summary: Chief executive officer (CEO) entrepreneurial orientation (EO) is an emerging topic within entrepreneurship research. This research offers the attention-based view (ABV), and its principles of attentional focus, situated attention, and structural distribution of attention to explain how CEO EO fosters an innovative organizational workforce. We theorize that CEO EO predicts lower-level employee innovative behavior when their top managerial attentional focus on entrepreneurship is translated into high-performance work practices, which structurally distribute their strategic attention throughout the organization and promotes a situated attentional condition for employees emphasizing innovation. Moreover, we consider (a) human-resource management (HRM) perceived importance and (b) departmental functional foci as boundary conditions. Using a multi-level and multi-source sample from 152 Chinese firms, results support our arguments.

Managerial Summary: Employee innovative behavior is critical to firm growth, renewal, and sustainability. By collecting and analyzing data on 2745 employees nested in 586 departments nested in 152 organizations, we examine the critical role that human resource management (HRM) systems and high-performance work practices may play in translating a CEOs emphasis on entrepreneurial activity into employee innovative behavior. We offer and observe



support for an attention-based perspective that HRM systems serve as “attentional structures” which channel CEO preferences to lower-organizational levels. Our results further indicate that an embrace of HRM as a strategic priority among top managers, as well as whether employees are in “boundary spanning” input–output areas of the business, to be meaningful factors which strengthen the effect of HRM-based attentional structures upon employee innovative behavior.

KEYWORDS

attention-based view, CEO entrepreneurial orientation, employee innovative behavior, human resource management importance, human resource management system

1 | INTRODUCTION

Recently, in the field of entrepreneurial orientation (EO) research, the chief executive officer's (CEO) EO has attracted attention from scholars (Keil et al., 2017; Liu & Xi, 2022; Zhang et al., 2021). Based on the attention-based view (ABV; Ocasio, 1997), CEO EO is defined as a CEO's consistent, observable attention and commitment to entrepreneurial activities and behaviors (Keil et al., 2017; Liu & Xi, 2022). CEO EO captures an entrepreneurial top management style (Wales et al., 2020), with the potential to shape how employees observe and respond to stimuli (Chirico et al., 2011).

EO is presumed to manifest within organizations such that a commitment to new ideas and experimentation pervades the organization at all hierarchical levels (Liu & Xi, 2022; Wales et al., 2011; Wiklund & Shepherd, 2011). Wales et al. (2020) posit CEO EO as capturing an entrepreneurial top management style that influences organization choices and drives a firm's innovative behavior and new entries. However, development of the theoretical linkage between CEO EO and employee innovative behavior remains an open question. Drawing on the ABV, and its three principles of attentional focus, situated attention, and structural distribution of attention as a theoretical framework, this study helps us understand and explain how CEO EO fosters a robust organizational embrace of employee innovative behavior, a phenomenon linked to a myriad of positive firm outcomes including growth, renewal, and sustainability (Rauch et al., 2009).

Although scholars have adopted an ABV (Ocasio, 1997, 2011) to explain what CEO EO captures, our theoretical understanding of managerial attention within EO research is presently incomplete and insufficient as it tends to focus only on the first principle of ABV. For instance, CEO EO, as a focus of attention on entrepreneurship, has been linked to firm performance (e.g., Keil et al., 2017; Liu et al., 2021) and strategic change (Wang et al., 2021). Similarly, Liu and Xi (2022) draw on ABV's attentional focus principle to offer an information-processing-based rationale for how CEOs may influence middle-managerial confidence and anxiety given that their managerial attention constitutes a limited, finite resource. While instructive, Ocasio et al. (2018) suggest moving beyond purely attentional-focus, information-processing-based logic when examining the ABV.

In this study, we demonstrate how all three principles of ABV work together to explain how CEO EO fosters an innovative organizational workforce. We begin by asserting a close linkage between CEO EO and ABV, for example, that CEO EO captures a CEO's focus of attention on entrepreneurship. This focus is manifest in terms of EO as a top management style, or “the top managerial goals, beliefs, logics, decisions, and communications which demonstrate

organizational commitment to *new entry*” (Wales et al., 2020, p. 644). To examine CEO EO is to consider the implications of their focus of attention on entrepreneurship, including what CEOs willingly emphasize within their communications (Keil et al., 2017) and decisions (Liu & Xi, 2022). Thus, taking attentional focus as definitive of CEO EO (ABVs 1st principle), in this research we consider the implications of CEO EO for an organization's structural distribution of attention (ABVs 3rd principle) and the situated attention of its employees (ABVs 2nd principle).

Reflecting the focus of attention of an organization's chief strategist, CEO EO can be regarded as a critical driver of the dominant logic of an organization (Covin & Lumpkin, 2011; Keil et al., 2017; Wales et al., 2020) with the potential to (a) drive the prevailing cognitive and attentional schema among managers and employees (Liu & Xi, 2022; Wales et al., 2020) and (b) focus employee efforts along an innovative path (Bettis & Prahalad, 1995). However, despite prevailing assumptions that EO manifests as a homogeneous strategic orientation (Wales et al., 2011), a shared, consistent attention to innovation throughout an organization is not a given due to differences in organizational roles, responsibilities, and hierarchies (Floyd & Lane, 2000). Lower-level employees often have different priorities which contribute to their cognition and behavior diverging from the expectations of organizational leadership (Kemelgor, 2002; Wales, 2016). Whether and how CEO EO promotes a shared attentional and behavior schema among lower-level employees, consistent with an organizational dominant logic toward entrepreneurial activity, is an important and urgent question that needs to be explored. This is because the presence of a holistic, firm-level entrepreneurial strategic orientation which spans CEO, TMT, and lower-level employee behavior is predicated upon an embrace of innovative, entrepreneurial thoughts, dialogue, and action throughout the organization.

In this research, we offer several contributions to ongoing conversations within literature. Our first contribution lies in demonstrating how additional principles of the ABV beyond its first assertion are necessary for a more complete understanding of CEO EO and its organizational implications. We assert that CEO EO captures the strategic focus of an organization's principal decision-maker on entrepreneurial activity which can be channeled through attentional structures, or “attentional coherence mechanisms” (Titus & Anderson, 2018, p. 499), to foster an entrepreneurial dominant logic within organizations (Covin & Lumpkin, 2011). As such, we theorize that CEO EO is fundamentally tied to attention, and that a more complete understanding of how this attention is directed through organizational elements and levers is critical to our understanding of how organizations become robust, entrepreneurial actors.

Our second contribution is to offer a more complete understanding of the theoretical puzzle of the vertical pervasiveness of EO by investigating whether, how, and when CEO EO reaches the nonmanagerial employee level, influences employee innovative behavior, and strengthens their firms' entrepreneurial dominant logic (Covin & Wales, 2019). Employee innovative behavior is defined as “an employee's intentional introduction or application of new ideas, products, processes, and procedures to his or her work role, work unit, or organization” (Yuan & Woodman, 2010, p. 324). The ABV holds that the structured distribution of the decision-makers' focus of attention depends on formal or informal procedural and communication channels (Ocasio, 1997, 2011). We argue that the innovative behavior of employees will align with CEO EO and reinforce an entrepreneurial dominant logic when the role expectations of employees are codified within attentional structures, for example, organizational systems, policies, and structures. Thus, we first explore a potentially critical mechanism—human resource management (HRM) systems characterized by high-performance work practices (HPWPs)—which CEOs with EO may create and employ as attentional structures within their organizations to situate employee attention on innovation.

Third, we examine boundary conditions which recognize that employees focus of attention is influenced by the characteristics of the situations or contexts they face (Ocasio, 1997, 2011). HRM importance and departmental functional area are theoretically linked because they may bound the vertical pervasiveness of CEO EO—one due to attitudes toward HRM which shape managerial attentional focus and the other due to the varying priorities of employee functional areas. Thus, we explore two specific contingencies that may foster the conditions necessary to regulate and channel the vertical pervasiveness of CEO EO to employee innovative behavior through HPWPs. We posit that CEOs with EO are more likely to design and adopt HPWPs as attention structures when HRM is embraced among top managers as a strategic business function (Lee, 2021). Considering “downstream” effects, we argue that



HPWP-based attention structures motivate employees to engage in innovative behaviors more so when their department is responsible for input and/or output functions of the organization. As opposed to throughput functions, which focus on improving the operational efficiency of an organization, input- and output-oriented functions focus on organizational boundary-spanning value-creation activities and communication roles (Bunderson, 2003; Datta & Rajagopalan, 1998; Greve et al., 2015; Liu & Xi, 2022). Taken together, Figure 1 presents our overall conceptual model.

2 | THEORETICAL FOUNDATION AND HYPOTHESES

The ABV rests upon three principles which together describe how organizations distribute and regulate the attention of their members (a) attentional focus, (b) situated attention, and (c) structural distribution of attention (Ocasio, 1997). Past research within entrepreneurship and strategy has generally emphasized the first principle of ABV and the limits of that individual decision-makers cognitive ability to pay attention to different competing initiatives (Ocasio et al., 2018). However, within organizations, top managerial attention can also be structurally distributed, for example, regulated and channeled through economic and social structures into “the activities, communications, and procedures which constitute the situational context of decision-making within an organization” (Ocasio, 1997, p. 189). Moreover, the situational context of decision-making, for example, the second principle or premise of the ABV, explains that what organizational decision-makers attend to depends upon their socio-cognitive context.

Therefore, within organizations the interconnectedness of ABV's principles can be illuminated in terms of top managers attentional focus, most notably CEOs (Chatterjee & Hambrick, 2007), influencing the creation of attentional structures that situate the attention of lower-level members toward certain areas and activities (Titus Jr & Anderson, 2018). In this vein, ABV offers a theoretical mechanism for explaining how organizational dominant logics in which “organizational attention is only focused on data deemed relevant to by the dominant logic” (Bettis & Prahalad, 1995, p. 7) are created and reinforced. When a dominant logic focused on entrepreneurship, experimentation with new ideas, and innovation develops, scholars characterize the organization as having manifested a robust, entrepreneurially oriented-strategic orientation (Covin & Lumpkin, 2011; Covin & Wales, 2019; Wales, 2016). In the proceeding sections, a theoretically derived role of attentional structures is brought to light which, if overlooked, would lead to imprecise predictions about whether and when a CEO's strategic focus on entrepreneurship will translate into an organizational embrace of innovation.

2.1 | CEO entrepreneurial orientation and employee innovative behavior

For many decades, scholars have explored how EO promotes firm performance, and widely supported a positive relationship (Anderson et al., 2022; Putniņš & Sauka, 2020; Rauch et al., 2009). As the field has evolved, scholars have

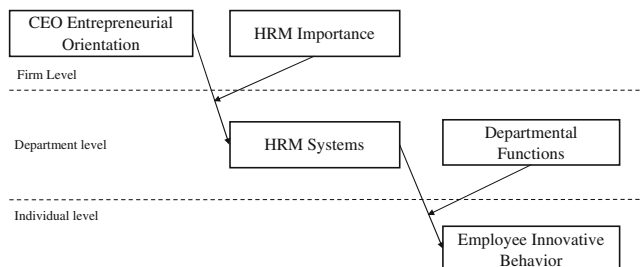


FIGURE 1 Research framework.

increasingly drawn attention to an emerging topic within the EO literature, namely the CEO's EO and its organizational consequences (Keil et al., 2017; Liu & Xi, 2022; Lumpkin & Pidduck, 2021; Wang et al., 2021; Zhang et al., 2021). CEO EO is usually described in terms of a CEO's consistent pattern of attention to and focus on entrepreneurial activities and behaviors (Keil et al., 2017; Ocasio, 1997). Following these studies, it is reasonable to speculate that entrepreneurially oriented CEOs cognitively differ from those with general qualities, particularly in their entrepreneurial cognitions. That is, high EO CEOs distinguish themselves from their peers by being more likely to eschew steady returns in favor of exploring new growth avenues and focusing their attention on entrepreneurial issues which demand embracing uncertainty, experimentation, and a highly variable distribution of outcomes (Wiklund & Shepherd, 2011).

Drawing on the ABV (Ocasio, 1997, 2011), researchers have argued that by paying consistent attention to entrepreneurship, CEOs demonstrate their strategic intent and encourage organizational members to engage in new value-creating activities and behaviors, thereby promoting key results desired by the organization such as increased firm performance (Liu & Xi, 2022) and market valuation (Keil et al., 2017). However, scholars have rarely focused on frontline employees' reactions to CEO EO (Liu et al., 2021; Wales, 2016; Wiklund & Shepherd, 2011; Zahra, 1993). Of note, employees play a pivotal role in implementing the strategic decision-making of top managers (Floyd & Lane, 2000), entrepreneurial activities and acts (Covin & Wales, 2019), and recognizing new opportunities (Mintzberg & Waters, 1985). Employees' innovative behaviors and engagement in value-creating entrepreneurial behaviors is therefore desirable by CEOs who have a high commitment to EO (Keil et al., 2017; Liu & Xi, 2022). Thus, we expect that CEOs with greater EO will seek to promote a positive impact on employee innovative behavior and provide supportive arguments based on attention-based theory and conceptualizations of EO as a vertically pervasive phenomenon.

To better identify changes in competitive environments and make effective decisions, CEOs need to focus their limited attention on what they perceive are key, strategic issues, to ensure an efficient utilization of organizational resources, understand signs of emerging threats, and develop successful strategies in competitive markets (Hambrick et al., 1996; McMullen et al., 2009; Ocasio, 1997). Thus, we begin by showcasing the first principle of the ABV. When CEOs devote their limited attention to entrepreneurial activities, for example, CEO EO, they demonstrate what they value to organizational members (Keil et al., 2017). In this vein, when CEOs focus their limited attention on specific firm issues, they are likely to receive more attention and participation from organizational members (Ocasio, 2011; Tushman & Rosenkopf, 1996). Thus, CEOs who demonstrate a clear emphasis on EO in matters they attend to may stimulate employees' innovative behaviors by suggesting their criticality to the firms' strategic direction and priorities (Zhao et al., 2011). Through employee observation, CEO EO itself may therefore socially influence the attention of lower-level employees, and thereby offer a directly situated attentional condition (ABV's second principle).

In this vein, a CEO's attention to and emphasis on entrepreneurial activities and behaviors may situate the attention of employees toward new ideas, motivating them to invest more time and energy into innovation activities and opportunities. A CEO's attention, emphasis, and openness to entrepreneurial behaviors and activities are likely to give salience and legitimacy to innovation-focused initiatives, investments, and practices across various levels within the organization. The CEO's consistent attention to entrepreneurship is likely to vertically convey an expectation throughout the organization that innovative behaviors are expected (Geletkanycz & Hambrick, 1997; Keil et al., 2017; Liu & Xi, 2022). When employees perceive such expectations, they are more likely to engage in innovative behaviors. Therefore, we propose:

Hypothesis 1. *CEO EO is positively related to employee innovative behavior.*

2.2 | The mediating effect of HRM systems

Although we expect that entrepreneurial-oriented CEOs are able to inspire innovative behaviors among their employees, the influence may be limited due to infrequent and diffuse communications between frontline employees



and the CEO which is often the result of management hierarchy within an organization (Xi et al., 2017). In other words, the CEO's consistent attention and commitment to entrepreneurship is not always directly observable and does not easily trickle down vertically to nonmanagerial employees. As such, the influence of CEO EO may be indirect.

From an organizational perspective, decision-makers utilize formal policies, systems, and/or practices to form rules of exchange or establish norms and standards of action among its members (Emerson, 1976; Shaw et al., 2009), and to guide and regulate the behavior of members toward organizational goals (North, 1990; Thornton & Ocasio, 1999). Therefore, we extend our theoretical model to incorporate the third principle of the ABV, for example, that within organizations decision-makers may create attention structures to channel and distribute their preferences throughout the company and, in turn, situate members attention on issues of relevance to organizational leadership (Ocasio, 1997).

The literature has largely supported that a firm's HRM system, defined as "a group of separate but interconnected human resource practices designed to enhance employees' skills and effort" (Takeuchi et al., 2007, p. 1069), is an effective way to contribute to organizational-level and individual-level outcomes by enhancing employees' abilities, motivations and opportunities within particular areas (Jiang et al., 2012; Paauwe & Boselie, 2003). Well-designed HRM systems are characterized by HPWPs which are employee skill-, motivation-, and opportunity-enhancing (Xi et al., 2022). These organizational systems support employee engagement through internal career opportunities, clear job descriptions, training opportunities, results oriented appraisals, employee security, participation, and profit sharing (Delery & Doty, 1996), and help convey organizational strategy, attention, and goals (Boada-Cuerva et al., 2019).

HRM systems offer a strategic lever (Boada-Cuerva et al., 2019; Bowen & Ostroff, 2004; Xi et al., 2021), that may channel and regulate the attention of organizational members along particular directions of competitive relevance to the firm. That is, HRM systems describe the common behavioral expectations among organizational members based on various practices, rules, and norms, designed to convey what the organization rewards and punishes and define what excellence resembles (Emerson, 1976; Jackson & Schuler, 1992; Jiang et al., 2012; Shaw et al., 2009). The logic of strategic HRM holds that how an organization manages its employees is a key function for business organizations to achieve competitive advantages and strategic goals (Lee, 2021), and the power of HRM systems depends on their ability to guide and regulate the attitudes and behaviors of organizational members through effective HPWPS (Bévort & Poufelt, 2015; Boxall & Purcell, 2011; Jackson et al., 2014).

As a strategic lever available to entrepreneurial CEOs, HPWPs may help motivate employee innovative behaviors (Escribá-Carda et al., 2017; Jiang et al., 2012; Lee et al., 2019). Specifically, training and development practices increase employees' knowledge, skills, and abilities (Combs et al., 2006), motivate them to exchange knowledge and analyze problems in a different way (Escribá-Carda et al., 2017), and help them better handle more complex job tasks and responsibilities (Amabile et al., 2004; Zhou & Hoever, 2014). Participation and empowerment practices create a suitable situation for employees to feel comfortable generating new ideas (Escribá-Carda et al., 2017), and afford critical autonomy when innovating (Volmer et al., 2012). Reward systems or profit-sharing practices can also play an important role in stimulating employees' innovative behaviors (Kuratko et al., 1999). Internal career opportunities and training promote employees' intrinsic motivation (Boxall et al., 2015), which may lead to more time and energy spent exploring novel information, understanding problems, and generating creative solutions (Ryan & Deci, 2000).

Drawing on attention-based theory (Ocasio, 1997, 2011), we expect that CEOs with EO may design and build HRM systems which function as attention structures designed to promote attentional coherence and the emergence of an entrepreneurial dominant logic through the organization. That is, CEOs have a decisive influence on the formulation of HRM philosophies, goals, policies, and practices (Hambrick, 2007; Steffensen et al., 2019), with the potential to trigger employees' innovative behavior. These policies promote attentional coherence throughout the organization by motivating employees to strive for high-performance, including defining in what areas job performance is most valued, and how employees advance within the organization. Research supports that HRM systems are strongly influenced by top managers (Boada-Cuerva et al., 2019; Steffensen et al., 2019), and perceived and experienced by

organizational members (Nishii & Wright, 2008; Xi et al., 2021). In this vein, the EO literature asserts that when EO is manifest as a top management style, it has the potential to directly shape the development of organizational attributes, including how processes, routines, structure, culture, and so forth are configured to support and foster innovative new entries (Wales et al., 2020). In other words, HRM systems may be heavily influenced by top management style (Wales et al., 2020), especially a CEOs entrepreneurial management style, which can be transmitted and codified within the organization as a set of HR norms and expectations.

Thus, HRM systems and HPWPs may serve as attention structures that channel CEOs' attention, emphasis, and openness to entrepreneurial activities throughout their organizations, thereby promoting employee innovative behavior. CEOs with different levels of EO are likely to differ in their willingness to design and adopt HPWPs as attention structures to channel their consistent attention and commitment to entrepreneurship. Compared with CEOs with low EO, CEOs with high EO have higher tolerance for innovation and entrepreneurial risks and expect their consistent attention and commitment to entrepreneurship to pervade the whole organization, forming a shared cognitive and attentional pattern among organizational members aimed at achieving excess risk returns. Thus, we argue that high EO CEOs may be more willing to proactively design and adopt HPWPs to serve as attention structures within their organizations than their low EO peers. In light of the above discussion, we hypothesize:

Hypothesis 2. *Human resource management systems and work practices mediate the positive relationship between CEO EO and employee innovative behavior.*

2.3 | The moderating effect of HRM importance

As mentioned, the second principle of ABV holds that what decision-makers focus on and what they allocate their attention to depends on their particular context or situation, referred to as situated attention (Ocasio, 1997). This suggests that a decision-makers socio-cognitive context is particularly salient to where and how they direct their attention. To understand when entrepreneurial CEOs design and adopt HRM systems as attention structures within their organizations to situate employee attention toward innovation, it is informative to consider when they will be more willing to proactively create such attention structures within their companies.

As senior leaders responsible for corporate strategy, the top management teams (TMTs) specific cognition and judgment concerning the importance of HRM as a strategic organizational function critical to the firm's mission is likely to have wide-reaching effects (Lee, 2021), including potentially moderating an organization's structural distribution of attention through HRM systems and HPWPs within an organization. The more organizational leadership pays attention to HRM systems and believes in their potential, the greater the role that these systems should play in employee behavior (Budhwar & Budhwar, 2000; Lee, 2021; Ocasio, 1997). Thus, as cognitive attitudes towards HRM may shape attentional focus within this area, we focus on HRM importance defined as the status of HRM in an organization relative to other functional departments for the business development of the firm (Wei & Lau, 2005). If the top management of a company believes that HRM is important and strategic to the future development and growth of the company, the company may devote more effort towards its strategic usage (Lee, 2021; Ocasio, 1997; Wei & Lau, 2005). Conversely, if HRM issues are not a priority of the company's top managers, they are likely to devote less of their attention to these issues. As such, we argue that HRM importance functions as a socio-cognitive condition that regulates how effectively a CEO's commitment to entrepreneurship is channeled through HPWPs. That is, greater perceived strategic importance of attention to HRM and work practices for business development is likely to amplify the effect of a CEO's attention to entrepreneurial activity upon the adoption of supportive HPWPs that create the situational conditions necessary for employee innovation.

Moreover, beyond socio-cognitive reasons, the often discussed first principle of ABV holds that decision-makers' attention is limited (Ocasio, 1997, 2011). Since top managers control and allocate an organization's limited resources when strategy-making (Tsui, 1987), the more attention that certain stimuli receive, the more support and



resources they are allocated. The more important HRM is, the more likely decision-makers will channel their limited attention toward HRM issues, give priority to HRM issues, and invest resources into HPWPs (Lee, 2021; Ocasio, 1997; Wei & Lau, 2005). When top managers consider HR as, or more, strategically relevant than other functional areas, they (e.g., the CEO and her leadership team), will budget more resources to HRM systems and support the creation of HPWPs (Mitsuhashi et al., 2000; Wei & Lau, 2005). Increased resource commitment towards HRM further enables the potential role of HRM systems and practices in promoting employee innovative behavior. In light of the above discussion, and drawing upon the ABV, we propose:

Hypothesis 3. *HRM importance among top managers moderates the indirect effect of CEO EO on employee innovative behavior via HRM systems such that the indirect effect will be stronger when HRM importance is higher than when it is lower.*

2.4 | The moderating effect of departmental functions

An important situational factor that shapes and channels the focus of organization members attention is organizational structure (Ocasio, 1997; Titus Jr & Anderson, 2018). As discussed, the third principle of ABV suggests that organizational elements may distribute decision-makers' attention throughout an organization. However, members in different departmental functions are often responsible for different issues, or different aspects of an issue, in the organization (Joseph & Ocasio, 2012; Kleinknecht et al., 2020; Ocasio & Joseph, 2005). As such, when CEOs attention toward entrepreneurship is distributed across functional departments different socio-cognitive lenses arise through which organizational members interpret the attention of top managers. That is, due to the different task environmental characteristics of various departmental business units within an organization, employee receptiveness to focusing their attention on entrepreneurially oriented initiatives is likely to differ (Covin & Wales, 2019; Wales et al., 2011). Thus, employees focus on attention is likely not homogeneously influenced within an organization due to socio-cognitive differences across departmental functions (Joseph & Ocasio, 2012; Kleinknecht et al., 2020), and certain departmental functions may be less impressionable to their CEOs focus of attention on entrepreneurship.

In this vein, we conceptualize organizational departmental functions as broadly divisible into two categories: throughput-oriented functions, which focus on improving operational efficiency of the organization largely within its boundaries, and input-output-oriented functions, which focus on productive activities that cross organizational boundaries between the organization and external entities (Bunderson, 2003; Datta & Rajagopalan, 1998; Greve et al., 2015). Throughput-oriented functions include activities such as production, accounting, law, human resources, audit, quality test, and operations. The core purpose of these functional activities is to achieve efficiency improvements in the process of value creation within the organization (Greve et al., 2015). Input-output-oriented functions incorporate product R&D, marketing, sales, logistics, supply chain, and communication roles. These activities within the organization aim to enable and enhance opportunities for organizational expansion (Greve et al., 2015).

In throughput-oriented functions, the core needs of the organization members' work are to develop job-specific skills and information processing capabilities to support the improvement of the organization's operational efficiency (Chaganti & Sambharya, 1987; Datta & Guthrie, 1994; Greve et al., 2015; Hambrick & Mason, 1984). In other words, the main requirement of employees in these functions is not innovation, but to improve the professionalism and proficiency of their work, so as to improve overall work efficiency. Although entrepreneurial CEOs may create attention structures that expect and support employee innovation, this does not imply that all employees, such as those in throughput-oriented functions need to, or in some instances (i.e., accounting), even should aspire to high levels of innovation.

On the contrary, in input- and output-oriented functions, the core requirement of organizational members' work is to pursue external opportunities and innovative product, market, and business model adaptations (Greve et al., 2015; Hambrick & Mason, 1984). In addition, the literature on boundary-spanning employees in organizations argues that these individuals need to master a combination of skills and innovative approaches (Tushman &

Scanlan, 1981), which suggests that they will take greater direction from their CEOs EO and benefit more strongly from HRM systems. Thus, when entrepreneurial-oriented CEOs design and build HRM systems as attention structures to channel their attention and commitment to entrepreneurship to employees and facilitate employee attention to innovation, we posit that employees belonging to input–output-oriented functions will be more likely to display innovative behaviors than those who are in throughput-oriented functions. Therefore, we hypothesize:

Hypothesis 4. *Departmental functional area moderates the indirect effect of CEO EO on employee innovative behavior via HRM systems such that the indirect effect is stronger when organizational employees belong to departments responsible for input–output-oriented functions rather than throughput-oriented functions.*

2.5 | Indirect relationship between CEO EO and employee innovative behavior

Taken together, our hypotheses suggest an indirect relationship between CEO EO and employee innovative behavior via HRM systems. CEOs with EO, seeking to improve their employees' innovative behaviors, support HRM systems to distribute their focus on entrepreneurship throughout the organization. Moreover, when HRM is embraced and attended to as an important strategic priority among top managers (Lee, 2021), CEO EO is more likely to prompt an increased design and adoption of HRM systems as attention structures to situate greater employee attention toward innovation. Further, we expect that the positive effect of HRM systems and employee innovative behavior will be stronger when employees are located within input- and output-oriented functional areas. Taken together, our theorizing thus suggests a conditional indirect effect of CEO EO on employee innovative behavior via HRM systems.

Hypothesis 5. *CEO EO has a conditional, positive indirect effect on employee innovative behavior via HRM systems. This indirect effect is strengthened when HRM importance among top managers is high, and employees belong to input–output-oriented functions.*

3 | MATERIALS AND METHODS

3.1 | Data collection and sample

In this study, we utilize a multi-source, multi-level research design to examine how CEO EO as a pervasive firm-level phenomenon influences employee innovative behavior through an embrace of HRM practices. Our sample mainly consists of chemical, electromechanical, new energy, medical, IT, and automotive manufacturing enterprises located at different Economic and Technological Development Zones, in China. Companies in these industries expect employees to demonstrate innovative behaviors, though are likely to differ in terms of their encouragement and success of achieving such employee behavior. With the help and support of development zone directors, 160 firms were identified and extended an invitation to participate in our survey from July 2020 to February 2021. To help assuage concerns regarding common method bias, we contacted at least two vice presidents, one human resource (HR) director, four department leaders, and approximately 20 employees within each firm to participate in our survey. For each firm we asked one vice president to rate their CEO's EO and another to rate the importance of HRM within the organization, we asked the HR director to rate the organization's overall embrace of HPWPs, the four department leaders to rate their departments usage of HPWPs, and the 20 employees randomly identified by four department leaders (five employees at each department) to describe their innovative behaviors.

In total, 160 firms returned the surveys. After deleting the observations with missing data, we obtained 152 firm-level, 586 department-level, and 2745 individual-level data. The final effective response rate for individuals



invited to participate was 85.78%. Of the 152 firms, 58 were high-tech; 72 were private; 105 were small- and medium-sized enterprises with less than 500 employees, and the average age was 17.28 years ($SD = 13.74$). Of the 152 rated CEOs, 132 were male with an average age between 46 and 50 years. Of the 586 departments, 377 were throughput functions, 209 were input/output functions, and with an average of 22 employees in each department ($SD = 28.13$). Of the 2745 employees, 1248 were male, between 30 and 35 years on average; they held a junior college degree on average, and had an average tenure of 5.16 years ($SD = 5.54$).

3.2 | Measures

3.2.1 | CEO entrepreneurial orientation

In this study, we employed the scale developed by Liu and Xi (2022) to measure CEO entrepreneurial orientation. The scale consists of three dimensions: innovativeness, proactiveness, and risk-taking. We asked vice presidents to rate CEO entrepreneurial orientation on a 7-point Likert scale from = “strongly disagree” to 7 = “strongly agree.” The Cronbach's coefficient of CEO entrepreneurial orientation was 0.957.

3.2.2 | Human resource management importance

We employed a four-item scale developed by Wei and Lau (2005) to measure the importance of HRM within the firm. Vice presidents were asked to rate HRM importance on a seven-point Likert scale ranging from = “strongly disagree” to 7 = “strongly agree.” The Cronbach's coefficient of HRM importance was 0.908.

3.2.3 | High-performance work practices

Data on our focal organizations' HPWPs adoption were collected from each firm's HR director as well as four department leaders. Specifically, department leaders were asked to evaluate their department's usage of HPWPs using a 23-item scale developed by Delery and Doty (1996). We asked the raters to indicate their agreement with each statement across 7 areas of HPWPs (the extent to which their departments HRM practices supported internal career opportunities, clear job descriptions, training, results-oriented appraisals, employee security, participation, and profit sharing) along a 7-point Likert scale (1 = strongly disagree; 7 = strongly agree), and the Cronbach's coefficient of department leader-rated HPWPs usage was 0.935. HR directors were asked to evaluate their firm's embrace of HPWPs using 15 items from Xi et al. (2022). We asked the raters to indicate their agreement with each statement across three areas of HPWPs (the extent to which their HRM system is characterized by skill-, motivation-, and opportunity-enhancing practices) along a 7-point Likert scale (1 = strongly disagree; 7 = strongly agree). The Cronbach's coefficient of HR director-rated organizational HPWPs adoption was 0.913.

3.2.4 | Departmental functional area

According to the classification of Hambrick and Mason (1984), Greve et al. (2015), and Liu and Xi (2022), the study divided the functional contribution of each department to the organizations mission into two distinct categorical “orientations”: input/output functions that emphasize growth and the development of product-market opportunities such as production innovation, marketing, sales, and R&D, and throughput functions which aim to improve an organizations efficiency at the transformation and new value creation process such as production, operations, human

resources, law, and accounting. Employees belonging to departments responsible for input/output functions were labeled as “1,” and throughput functions as “0.”

3.2.5 | Employee innovative behavior

In this study, we used a six-item scale developed by Scott and Bruce (1994) to measure employee innovative behavior. Employees were asked to indicate their innovative behaviors on a scale ranging from 1 = “never” to 7 = “always.” The Cronbach's coefficient of innovative behavior was 0.930.

All the scale items of measures can be seen in Appendix 1.

3.2.6 | Control variables

In this study, we controlled for four types of variables due to the nested structure of our data. First, we controlled for employee gender (1 = male, 0 = female), educational level (1 = high school or below, 2 = junior college, 3 = undergraduate degree, 4 = postgraduate or above), and tenure (by months). Second, we controlled for department size. Third, we controlled for CEOs' demographic variables including gender (1 = male, 0 = female) and education (1 = high school or below, 2 = junior college, 3 = undergraduate degree, 4 = postgraduate, 5 = PhD). Third, we controlled for firm characteristics including firm ownership (1 = private firm, 0 = others), firm industry (1 = high-tech firm, 0 = others), firm size, and established years.

3.3 | Analytical approach

Given that we collected CEO EO and HRM importance at the firm level, HPWPs at the firm and departmental level, departmental functional area at the department level, and employee innovative behavior at the individual level, we employed hierarchical linear modeling (HLM; Bryk & Raudenbush, 1992) to test our research hypotheses and account for the nested structure of our data (employees nested in departments, nested in organizations).

Before conducting our primary analyses, we first ran a null model with HPWPs to make sure that meaningful between-firm variance was detected. The results indicated that HPWPs had significant between-firm variance [$\chi^2_{(151)} = 325.203, p < 0.001, ICC(1) = 0.23$]. We then ran a null model with employee innovative behavior to make sure that both meaningful between-firm variance and between-department variance existed. The results showed that employee innovative behavior had significant between-firm variance [$\chi^2_{(151)} = 318.49, p < 0.001, ICC(1) = 0.09$] and between-department variance [$\chi^2_{(434)} = 877.28, p < 0.001, ICC(1) = 0.16$], indicating that HLM was appropriate for testing our hypotheses.

In addition, to confirm the distinctiveness of CEO EO, HRM importance, HPWPs, and employee innovative behavior, we conduct a multilevel CFA using Mplus 8.0. Following Schumacker and Lomax (1996), we found that the results showed a relatively good fit for the four-factor model [$\chi^2 = 1828.36, df = 269, p = 0.000, RMSEA = 0.046, CFI = 0.908, TLI = 0.892, SRMR(Within) = 0.037, SRMR (Between L2) = 0.055, SRMR(Between L3) = 0.056$].

4 | RESULTS

Table 1 provides means, standard deviations, and correlations for our focal variables. To avoid common method variance (Lindell & Whitney, 2001), we used department leader-rated HPWPs in our primary analyses. We then tested the robustness of our empirical findings by using the alternative, HR director-rated HPWPs as the mediator.



TABLE 1 Descriptive statistics and correlations among the variables.

Variables	Mean	S.D.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1. Employee gender	0.455	0.500																
2. Employee age	3.891	1.448	0.072**															
3. Employee education	2.508	0.786	0.026	-0.188**														
4. Employee tenure	61.918	66.442	0.065**	0.556**	-0.097**													
5. Department size	22.651	28.538	0.086**	0.046*	-0.089**	0.049*												
6. CEO gender	0.871	0.335	0.020	0.073**	-0.014	0.044*	0.027											
7. CEO education	3.811	1.234	0.019	-0.079**	0.264**	-0.021	-0.018	0.009										
8. Firm ownership	0.452	0.498	-0.067**	-0.156**	-0.023	-0.131**	-0.129**	-0.074**	0.059**									
9. Firm industry	0.389	0.488	0.056**	-0.079**	0.118**	-0.079**	0.110**	0.089**	0.139**	0.149**								
10. Firm size	3.346	1.303	0.018	0.034	0.135**	0.107**	0.147**	0.214**	0.058**	-0.212**	-0.006							
11. Firm age	17.538	13.657	-0.017	0.168**	0.003	0.236**	-0.020	0.040*	-0.101**	-0.043*	0.007	0.165**						
12. CEO EO	5.790	0.971	0.008	-0.096**	0.003	-0.142**	-0.032	-0.111**	0.040*	0.128**	0.123**	-0.055**	-0.222**					
13. HRM importance	5.807	1.014	0.050**	-0.001	0.017	-0.041*	0.008	-0.133**	0.005	0.031	-0.126**	0.052**	-0.035	0.391**				
14. HR-rated HPWPs	5.776	0.782	-0.053**	-0.127**	0.105**	-0.108**	-0.008	-0.004	0.117**	0.006	0.019	-0.034	-0.094**	0.160**	0.161**			
15. Department leader-rated HPWPs	5.493	0.942	0.044*	0.054**	0.044*	0.034	0.019	-0.051**	-0.024	-0.066**	0.006	0.011	-0.008	0.142**	0.119**	0.244**		
16. Input/output functions	0.359	0.480	0.072**	-0.068**	0.155**	-0.069**	-0.137**	-0.029	0.031	0.050**	0.017	-0.067**	-0.090**	0.112**	0.009	0.106**	0.019	
17. Employee innovative behavior	5.135	1.129	0.163**	0.017	0.072**	0.023	0.008	-0.065**	0.021	-0.043*	0.022	0.023	-0.013	0.116**	0.125**	0.117**	0.193**	0.043*

Note: N = 2745 at the individual level, N = 586 at the department level, and N = 152 at the firm level; department- and firm-level variables were de-aggregated to provide estimates of within-person relationships with the individual-level variables. Gender: 1 = male, 0 = female; Firm ownership: 1 = private firm, 0 = others; Firm industry: 1 = high-tech firm, 0 = others.
Abbreviations: CEO, chief executive officer; EO, entrepreneurial orientation; HPWPs, high-performance work practices; HRM, human resource management.
*p < 0.05; **p < 0.01.

4.1 | Hypothesis testing

Table 2 provides HLM3 regression results predicting employee innovative behavior. According to Table 2, CEO EO was positively related to employee innovative behavior ($\gamma = 0.137, p < 0.01$, Model 5) after accounting for control variables, thus supporting Hypothesis 1. After accounting for control variables, the results in Table 2 also suggest that HPWPs are positively related to employee innovative behavior ($\gamma = 0.180, p < 0.001$, Model 6).

Table 2 also shows the regression results predicting the use of HPWPs. In Model 2, CEO EO positively and significantly related to departmental HPWPs ($\gamma = 0.164, p < 0.01$). Model 7 indicates that, after controlling for CEO EO, the effect of departmental HPWPs on employee innovative behavior was still positive and significant ($\gamma = 0.170, p < 0.001$). We also estimated this indirect effect by employing a robust Monte Carlo simulation across 20,000 replications to obtain an estimate of the bias-correlated 95% confidence interval around this potential indirect relationship (Selig & Preacher, 2008). The results offer confirmation of a significant indirect effect (indirect effect = 0.030, 95%CI = [0.021, 0.043]), thereby supporting Hypothesis 2.

Results in Model 3 show that the extent to which an HRM system is perceived as important within an organization interacts with CEO EO to predict a department's use of HPWPs ($\gamma = 0.107, p < 0.05$). Following Preacher et al.'s (2010) recommendations, we conducted simple slope analyses to further probe our findings. Results (see Figure 2) revealed that the positive relationship between CEO EO and departmental use of HPWPs was stronger when the organizational importance and embrace of HRM was high ($\gamma = 0.257, p < 0.001$) than when it was low ($\gamma = 0.041$, n.s.). Furthermore, we conducted a Monto Carlo confidence interval for the conditional indirect effect of CEO EO on employee innovative behavior via HPWPs at high and low levels of HRM importance (Selig & Preacher, 2008). Results indicated that the indirect effect of CEO EO on employee innovative behavior via HPWPs was stronger when HRM importance was high (indirect effect = 0.038, 95% CI = [0.016, 0.061]) than that was low (indirect effect = 0.006, 95% CI = [-0.016, 0.028]). Thus, our Hypothesis 3 was supported.

Hypothesis 4 proposed the moderating effect of departmental functional area in the indirect effect of CEO EO on employee innovative behavior via HPWPs. Results of Model 8 in Table 3 suggested a significant interactive effect of HPWPs and departmental functional area on employee innovative behavior ($\gamma = 0.085, p < 0.01$). Results (see Figure 3) revealed that the positive relationship between HPWPs and employee innovative behavior was stronger when employees belong to input/output functions ($\gamma = 0.190, p < 0.001$) than when their focus was more throughput oriented ($\gamma = 0.108, p < 0.05$). Similarly, the Monto Carlo confidence intervals for indirect effect indicated that the indirect effect of CEO EO on employee innovative behavior via HPWPs was stronger when organizational employees were in input- and output-oriented functions (indirect effect = 0.035, 95%CI = [0.017, 0.039]) than that in throughput-oriented functions (indirect effect = 0.010, 95%CI = [0.005, 0.027]). Thus, our Hypothesis 4 was supported.

Hypothesis 5 proposed that the indirect effect of CEO EO predicting employee innovative behavior via HPWPs is moderated by organizational HRM importance and departmental functional area. To test these conditional indirect effects, we multiplied the coefficients for the simple slopes of CEO EO predicting departmental usage of HPWPs at varying levels of HRM importance (from Model 3 of Table 2) by the coefficients for the simple slopes of departmental HPWPs usage predicting employee innovative behavior (from Model 8 of Table 2) to obtain estimates for the conditional indirect effects. Table 3 presents the Monto Carlo confidence intervals for these conditional indirect effects (Selig & Preacher, 2008). Offering support for hypothesis 5, we find that the indirect effect of CEO EO as a predictor of employee innovative behavior via HPWPs was most positive and significant when HRM importance was high within the organization and employees belonged to input/output functions (indirect effect = 0.060, 95% CI = [0.018, 0.088]).

To confirm the robustness of the above results and address potential endogeneity concerns, we first conducted alternative analyses for all regression models displayed in Table 2 after removing all control variables and found similar results, and all hypotheses were supported. Second, considering issues around causal inference with EO research (e.g., firm performance, seeing Anderson et al., 2022), we ran all of our models after adding firm performance



TABLE 2 Results of hierarchical regression analysis for department HPWPs and employee innovative behavior.

Variables	Department HPWPs			Employee innovative behavior				
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Intercept	5.485***	5.483***	5.444***	4.892***	4.881***	4.917***	4.904***	4.923***
Employee gender (L1)				0.302***	0.302***	0.300***	0.300***	0.295***
Employee age (L1)				−0.004	−0.004	−0.007	−0.007	−0.007
Employee education (L1)				0.036	0.040	0.033	0.036	0.035
Employee tenure (L1)				0.001	0.001 ⁺	0.000	0.001	0.001 ⁺
Department size (L2)	0.000	0.001	0.000	−0.000	−0.000	−0.000	−0.000	−0.000
CEO gender (L3)	−0.093	−0.049	−0.041	−0.227*	−0.187 ⁺	−0.204*	−0.173 ⁺	−0.183*
CEO education (L3)	−0.036	−0.034	−0.030	0.001	0.002	0.007	0.007	−0.004
Firm ownership (L3)	−0.143	−0.174 ⁺	−0.184 ⁺	−0.079	−0.105	−0.055	−0.077	−0.084
Firm industry (L3)	0.042	−0.001	0.039	0.061	0.025	0.053	0.026	0.058
Firm size (L3)	−0.010	−0.011	−0.002	0.015	0.013	0.053	0.015	0.015
Firm age (L3)	−0.000	0.002	0.001	−0.001	0.001	0.016	0.000	0.000
Department HPWPs (L2)						0.180***	0.170***	0.149***
Input/output functions (L2)								0.011
Department HPWPs (L2) × Input/output functions (L2)								0.085**
CEO EO (L3)		0.164**	0.149*		0.137**		0.110*	0.073 ⁺
HRM importance (L3)			0.084 ⁺					0.082*
CEO EO (L3) × HRM importance (L3)			0.107*					−0.008
Pseudo R ²	0.00	0.016	0.028	0.035	0.048	0.067	0.075	0.084
Δ Pseudo R ²	0.00	0.016	0.012	0.035	0.013	0.032	0.027	0.009

Note: N = 2745 at the individual level (L1), N = 586 at the department level (L2), and N = 152 at the firm level (L3).
Abbreviations: CEO, chief executive officer; EO, entrepreneurial orientation; HPW5s, high-performance work practices; HRM, human resource management.
⁺p < 0.10;
*p < 0.05; **p < 0.01; ***p < 0.001.

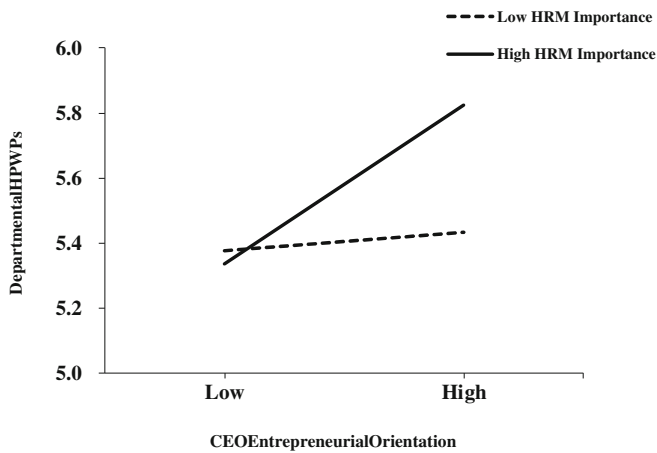


FIGURE 2 The moderating effect of HRM importance in the relationship between CEO entrepreneurial orientation and department HPWPs.

TABLE 3 Bootstrap estimation of Monte Carlo confidence intervals for conditional indirect effects.

Moderators		Indirect effect (CEO EO → HPWPs → Employee innovative behavior)	95% CI
HRM importance	Departmental functions		
Low (Mean – SD)	Throughput functions	0.003	[–0.013, 0.023]
	Input/output functions	0.010	[–0.021, 0.038]
High (Mean + SD)	Throughput functions	0.016	[0.006, 0.058]
	Input/output functions	0.060	[0.018, 0.088]

Abbreviations: CEO, chief executive officer; CI, confidence interval; EO, entrepreneurial orientation; HPWPs, high-performance work practices; HRM, human resource management.

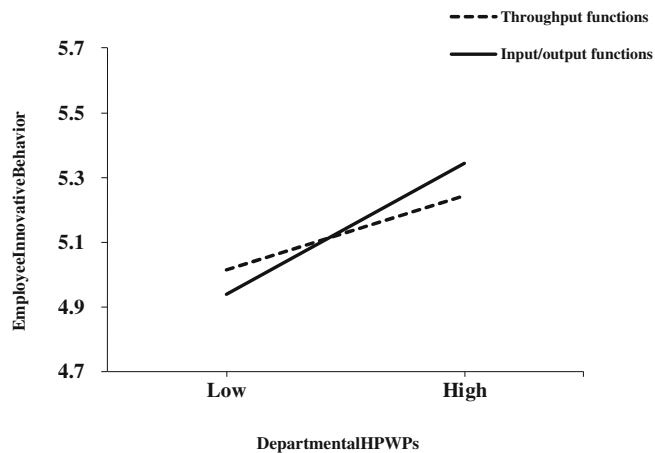


FIGURE 3 The moderating effect of departmental functions in the relationship between departmental HPWPs and employee innovative behavior.



(measured by seven items; Wang et al., 2003; $\alpha = 0.931$) as a new control variable to reduce the potential for omitted variable bias. Results still supported our hypotheses. Third, we used HR director-rated HPWPs adoption within the overall organization to test all proposed hypotheses. The supportive regression results can be found in Table A1.

5 | DISCUSSION

This study aims to examine the implications of CEO EO by elaborating upon how a CEO's attention, emphasis, and openness to entrepreneurial activities and behaviors may affect employee innovative behavior through HRM systems and work practices. Contributing to the growing stream of CEO EO literature, we explore a first-stage moderating effect of HRM importance on the relationship between CEO EO and HPWPs and a second-stage moderating effect of departmental functional area on the relationship between HPWPs and employee innovative behavior. Empirical data from 2745 employees nested in 586 departments nested in 152 organizations was collected to test our proposed model. In the sections below, we discuss the theoretical and practical implications of our findings, as well as future research opportunities.

5.1 | Theoretical implications

Looking first at the theoretical implications of our study, our research represents an important step towards completing the theoretical puzzle of how EO is manifest as an organizational strategic orientation and dominant logic embraced vertically from top-level management to individual employees, as well as horizontally across business units and functional areas (Covin & Lumpkin, 2011; Liu & Xi, 2022; Wales et al., 2011). In this research, a theoretically derived role of attentional structures is brought to light which, if overlooked, would lead to imprecise predictions about whether and when a CEO's EO will translate into an organization characterized by an innovative workforce that routinely introduces, experiments with, and applies new ideas, products, processes, and procedures (Wiklund & Shepherd, 2011).

Drawing on the ABV (Ocasio, 1997, 2011; Ocasio et al., 2018), organizational attentional coherence is promoted through the creation of attentional structures which distribute a CEOs focus on entrepreneurship throughout the organization, shaping the beliefs and behaviors of organizational members. Our study provides empirical support that a CEO's consistent attention and commitment to entrepreneurship influences employee innovative behavior via HRM systems and work practices. In doing so, our study responds to calls for a richer understanding of the cross-level facilitative conditions of organizational EO (Liu et al., 2021; Wales et al., 2020). Although studies have begun to explore the consequences of CEO EO, they have largely focused on outcomes at the firm level, such as firm strategic change (Wang et al., 2021), firm value creation (Keil et al., 2017), and firm performance (Liu & Xi, 2022). Our study contributes to a growing stream of multilevel research aimed at understanding the complex phenomena of organizational entrepreneurship (Kim et al., 2016; Liu & Xi, 2022).

Our research further deepens scholars' understanding of the mechanisms and mediators through which CEO EO shapes employee innovative behavior. Due to limited, infrequent, and diffuse communications between a CEO and its organization's frontline employees (Xi et al., 2017), the CEO's consistent attention and commitment to entrepreneurship is less likely to pervade vertically nonmanagerial employees and promote their innovative behavior. While past research has emphasized the criticality of understanding organizational leaderships focus of attention (e.g., the first principle of the ABV), our study draws attention to the criticality of identifying attentional structures which convey organizational leaderships focus of attention to subordinate members within the organization. As such, explorations and assessments of mechanisms and mediators through which CEO EO operates beyond HRM systems and practices are needed and encouraged (Covin & Wales, 2019; Wales, 2016; Wiklund & Shepherd, 2011).

While studies have adopted the ABV to explain what CEO EO is and how it affects outcomes (Keil et al., 2017; Liu & Xi, 2022), in most instances past conceptualizations have only drawn upon one or two of the three principles to explain the effects of managerial attention (Liu & Xi, 2022; Titus & Anderson, 2018; Zhang et al., 2021), thereby leaving our theoretical understanding of managerial attention within EO research insufficient to explain attentional congruence. Thus, our study contributes to a more comprehensive theoretical understanding of managerial attention within the strategic entrepreneurship literature through integrating the ABV with a richer, multi-level conceptualization of EO (Wales et al., 2020).

Beyond our contributions to the entrepreneurship literature, our research contributes to the human resource literature in three ways. First, drawing on the third principle of the ABV (Ocasio, 1997, 2011), we contribute to the implication of HRM systems and work practices in the entrepreneurial process by theorizing HPWPs as attentional structures. This attentional structure conveys CEOs' attention to and emphasis on entrepreneurship to frontline employees, and motivates and supports their engagement in innovative behaviors (Zotto & Gustafsson, 2008). As such, this research underscores the importance of HRM systems in linking a CEOs strategic orientation with the behavior of front-line employees operating in departments across the firm (Lee, 2021; Liu et al., 2021). Second, a greater understanding of the triggers of CEO decisions within HRM research is necessary (Steffensen et al., 2019). In this vein, our study advances the understanding of the roles of CEOs within the strategic HRM literature by indicating that CEOs with EO are more likely to adopt HPWPs. Third, we contribute to the ABV of strategic human resource management by demonstrating its applicability to CEOs with EO and, in particular, its utility in explaining the stronger relationship between CEO EO and employee innovative behavior via HPWPs (Lee, 2021; Wei & Lau, 2005).

5.2 | Practical implications

For organizations seeking to motivate employee innovative behavior, our research offers several key practical implications. First, in the vein of "lead by example," our research provides insight into how CEOs can, first and foremost, promote employee innovative behavior by demonstrating their personal attention, emphasis, and openness to entrepreneurial activities and behaviors. That is, this study finds that CEOs with EO enhance employee innovative behavior by showcasing their attention and codifying their support for employee innovative activity within a supportive organizational environment, for example, HRMs. In this vein, we encourage CEOs to publicly demonstrate their attention to entrepreneurial behaviors and strategies such as within speeches, media appearances, earnings calls, and letters to shareholders, and codify their support within HRM-centric policies and practices that further encourage innovative employee behavior. In a CEO's pursuit of innovation, it is of great importance to invest resources into HPWPs as these organizational practices demonstrate their willingness to "walk the talk" and provide a secure, nurturing organizational environment conducive to employee innovative behavior. Finally, given that HPWPs are particularly beneficial to increase employee innovative behavior for those who are in input/output functions, organizations may also benefit from designing their HPWPs in a manner which promotes employee attention towards supplier and/or customer issues. For instance, Amazon asks employees to write a "memo" detailing how a proposed initiative, even when internally focused, will directly influence the customer's experience in some manner.

5.3 | Limitations and future research directions

As with all research, our study has limitations that provide opportunities for future research. First, the data, while extensive, does not permit determination of a cause-effect relationship as implied in our model (Figure 1). In general, causal sequences are difficult to ascertain within organizational relationships. In addition, in this study, although we have employed two techniques and approaches including adding appropriate control variables to rule out potential omitted variables and performing robustness checks to detect and address endogeneity concerns (Busenbark



et al., 2022), testing for endogeneity in cross-sectional survey data can be challenging because cross-sectional data typically lacks the time dimension necessary to establish a temporal order of events. Thus, future studies might consider longitudinal experiments to better assess the proposed cause–effect relationship and employ additional approaches, such as the impact threshold of a confounding variable (Busenbark et al., 2022), to further address potential endogeneity concerns. Second, in this study, we pay attention to HPWPs as attention structures in linking CEO EO and employee innovative behavior. However, potential mediating mechanisms may extend beyond HPWPs. For example, an entrepreneurial culture demonstrates a pattern of shared values, or a collective mind that shapes organizational members' behavior and attitudes (Schein, 1992). Culture and values are strongly influenced by top management style (Wales et al., 2020) and can serve as attention structures that channel decision-makers' attention to desired issues (Rindova & Martins, 2018). Thus, entrepreneurially oriented CEOs may establish HPWPs as part of a broader entrepreneurial culture or system of values within the organization that promotes innovative behavior among employees. Other practices, artifacts, and aspects of entrepreneurial culture, values, or climate are also likely to be meaningful considerations (Kang et al., 2016).

In addition to the above limitations and research directions, our study also triggers some further research perspectives. First, what happens when CEOs are replaced (e.g., Grünh et al., 2017)? Is there inertia or change in the design and use of HRM systems and HPWPs? If there is change, in what ways and over what time period? Future research should further consider the dynamics of how CEO EO is converted into an innovative organizational workforce, and whether, when, and how this conversion might be constrained by the extant organizational conditions in place within the organization. Moreover, possible configurational relationships between CEO EO, organizational factors, and environmental considerations could be informative (Wales, 2016).

Second, there is a consistent emphasis on more explicitly incorporating theoretical perspectives into the EO research domain (Covin & Lumpkin, 2011; Wales et al., 2021; Wales et al., 2023). As an emerging consideration within EO research, the theories used to explain how CEO EO affects outcomes or consequences are still not abundant. Existing scholars, including this study, mainly employ attention-based theory to explain how CEO EO affects its outcomes. While a good start, future studies are encouraged to apply new theories from different perspectives to explain the impact of CEO EO.

Third, our study empirically supports the manifestation and pervasiveness of CEO EO within organizations vertically from top-level management to frontline employees, and horizontally across functional areas (Liu & Xi, 2022; Wales, 2016; Wales et al., 2020). However, EO's manifestation also has a temporal or dynamic perspective (Wales et al., 2011). In the current EO literature, there has been a lack of dynamic research on EO's manifestation. Thus, we encourage future research, on the one hand, to use a temporal perspective to address how and why the distribution of CEO EO across an organization may change over time, and on the other hand, to simultaneously integrate the vertical, horizontal, and temporal perspectives to investigate the pervasiveness of CEO EO within organizations. It is our hope that this study serves as a springboard upon which future research exploring CEO EO and its effects continues to build.

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APPENDIX 1: SCALE ITEMS

CEO Entrepreneurial Orientation (Liu & Xi, 2022, $\alpha = 0.957$)

Innovativeness

Our CEO emphasizes innovative solutions to problems.

Our CEO pays attention to innovation and updates of products and/or services.

Our CEO emphasizes investments in new technologies, methods, and procedures.

Proactiveness

In the face of market competition, our CEO emphasizes being proactive rather than reactive.

Our CEO emphasizes the proactive introduction of new products, services, technologies, and business models.

Our CEO emphasizes active identification and exploitation of new business opportunities by taking advantage of them.

Risk-taking

Our CEO is willing to take higher risks in pursuit of higher returns.

Our CEO has a high tolerance for future uncertainty and is not afraid to change.

Our CEO is willing to take the risk of failure when setting up new projects or ventures.

Human Resource Management Importance (Wei & Lau, 2005, $\alpha = 0.908$)

Top managers believe HRM is the key to the development of our business.

HR-related issues are always the priority that top managers consider in making decisions.

Relative to other managerial functions (such as marketing and finance), top managers pay more attention to HR issues.

HR department has an advantage over other functional departments in terms of budget.

Department manager-rated human resource management practices (Delery & Doty, 1996, $\alpha = 0.935$)

Internal career opportunities

Individuals in this job have clear career paths within the organization.

Individuals in this job have very rich future within this organization.

Employees' career aspirations within the company are known by their immediate supervisors.

Employees in this job who desire promotion have more than one potential position they could be promoted to.

Training

Extensive training programs are provided for individuals in this job.

Employees in this job will normally go through training programs every few years.



There are formal training programs to teach new hires the skills they need to perform their jobs.
Formal training programs are offered to employees in order to increase their promotability in this organization.

Results-oriented appraisals

Performance is more often measured with objective quantifiable results.
Performance appraisals are based on objective, quantifiable results.

Employment security

Employees in this job can expect to stay in the organizations for as long as they wish. It is very difficult to dismiss an employee in this job.
Job security is almost guaranteed to employees in this job. If the bank were facing economic problems, employees in this job would be the last to get cut.

Participation

Employees in this job are allowed to make many decisions.
Employees in this job are often asked by their supervisor to participate in decisions. Employees are provided the opportunity to suggest improvements in the way things are done. Superiors keep open communications with employees in this job.

Job descriptions

The duties of this job are clearly defined.
This job has an up-to-date job description.
The job description for this job contains all of the duties performed by individual employees.
The actual job duties are shaped more by a specific job description than by the employee.

Profit sharing

Individuals in this job receive bonuses based on the profit of the organization.

HR-rated human resource management practices (Xi et al., 2022, $\alpha = 0.913$)

Skill-enhancing

Our employees are administered an employment test (e.g., skills tests) prior to hiring.
Our employees are provided a formal job analysis to identify their position requirements (such as required knowledge, skills, or abilities).
Our employees receive continuous training, for example, continuous professional development.



Our employees receive formal performance appraisals to help them promote skill development and achieve better performance goals.

Motivation-enhancing

Our employees are offered training courses to help them obtain internal promotions.

Our employees receive formal individual performance appraisals.

Our employees have access to company incentive plans, profit-sharing plans, and/or gainsharing plans.

Our employees receive their performance appraisals which are used to determine their compensation.

Our company provides above-market wage levels to attract and retain employees.

Our company establishes a clear career path for employees.

Our employees are organized in work teams in performing a major part of their work roles.

Opportunity-enhancing

Our employees are included in a formal information-sharing program (e.g., a newsletter).

Our employees are asked to complete attitude surveys on a regular basis.

Our employees are encouraged to provide constructive suggestions to improve their work.

Our employees have access to a formal grievance procedure and/or complaint resolution system.

Employee innovative behavior (Scott & Bruce, 1994, $\alpha = 0.930$).

Searches out new technologies, processes, techniques, and/or product ideas.

Generates creative ideas.

Promotes and champions ideas to others.

Investigates and secures funds needed to implement new ideas.

Develops adequate plans and schedules for the implementation of new ideas.

Is innovative.



APPENDIX 2

TABLE A1 Results for firm HPWPs and employee innovative behavior.

Variables	Firm HPWPs (SPSS)			Employee innovative behavior (HLM3)							
	Model1	Model2	Model3	Model4	Model5	Model6	Model7	Model8			
Intercept	5.728	5.688	5.567	4.892***	4.881***	4.884***	4.873***	4.877***			
Employee gender (L1)				0.302***	0.302***	0.311***	0.310***	0.306***			
Employee age (L1)				−0.004	−0.004	−0.001	−0.001	−0.000			
Employee education (L1)				0.036	0.040	0.033	0.036	0.034			
Employee tenure (L1)				0.001	0.001+	0.001+	0.001+	0.001+			
Department size (L2)				−0.000	−0.000	−0.000	−0.000	−0.000			
CEO gender (L3)	0.022	0.058	0.075	−0.227*	−0.187+	0.235*	−0.201+	−0.179+			
CEO education (L3)	0.053	0.054	0.059	0.001	0.002	−0.009	−0.008	−0.007			
Firm ownership (L3)	0.034	0.006	−0.006	−0.079	−0.105	−0.078	−0.099	−0.096			
Firm industry (L3)	−0.017	−0.055	0.003	0.061	0.025	0.063	0.032	0.073			
Firm size (L3)	−0.032	−0.031	−0.021	0.015	0.013	0.020	0.018	0.014			
Firm age (L3)	−0.003	−0.002	−0.003	−0.001	0.001	−0.001	0.001	0.000			
Firm HPWPs (L3)						0.148***	0.130***	0.132***			
Input/output functions (L2)								0.013			
Firm HPWPs (L3) × Input/output functions (L2)								0.066*			
CEO EO (L3)		0.137*	0.106		0.137**		0.118**	0.085+			
HRM importance (L3)			0.133+					0.071+			
CEO EO (L3) × HRM importance (L3)			0.147*					−0.006			
(Pseudo) R ²	0.015	0.043	0.090	0.035	0.048	0.051	0.060	0.065			
Δ (Pseudo) R ²	0.015	0.028	0.048	0.035	0.013	0.016	0.012	0.005			

Note: N = 2745 at the individual level (L1), N = 586 at the department level (L2), and N = 152 at the firm level (L3).
Abbreviations: CEO, chief executive officer; EO, entrepreneurial orientation; HPWPs, high-performance work practices; HRM, human resource management.
+p < 0.10;
*p < 0.05; **p < 0.01; ***p < 0.001.